Customer Churn Analysis Report

This report presents the exploratory data analysis and data preparation steps to predict customer churn using various data sets. It includes data collection, data cleaning, and data visualization processes, providing a foundation for developing a predictive model.

Data Identification and Collection

The following data sets were identified and collected for predicting customer churn:

- Customer Demographics
- Transaction History
- Customer Service Interaction
- Online activity
- Churn status

These data sets provide comprehensive insights into customer behavior and engagement.

Exploratory Data Analysis

Exploratory data analysis revealed that **behavioral factors are stronger predictors of churn than demographics**. While age and gender distributions remain stable across groups, **customer service friction** emerged as a primary risk factor, with unresolved issues strongly correlating with churn. Interestingly, **financial metrics** like spending amount showed little variance between retained and churned customers, suggesting that high-value customers are just as likely to leave if unsatisfied.

Figure 1: Churn rate by gender

- **Insights**: There is slight gender disparity in retention. Male customers show a marginally higher churn rate (slightly above 20%) compared to female (just below 20%), suggest male might be slightly more at risk.

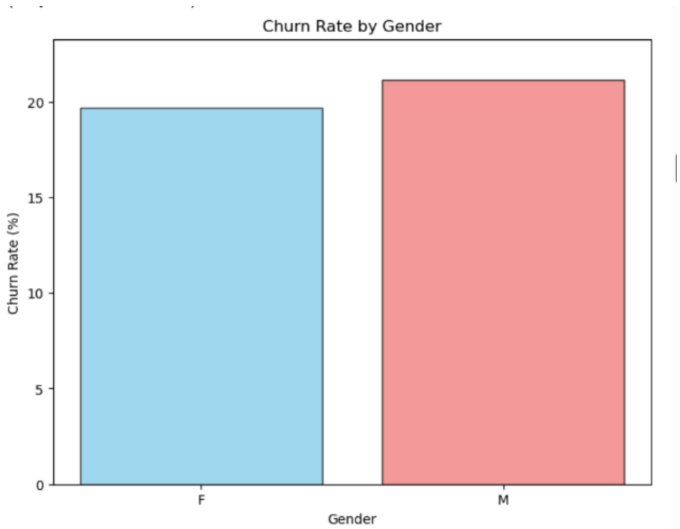


Figure 2: Customer Demographic

- Insights: The customer base is well-balanced. Gender distribution is nearly equal (approx..50/50), and age distribution are consistent across different income levels (Low, Medium, High). This stability implies that churn is likely driven by behavioral factors (like service issues) rather than inherent demographic bias.

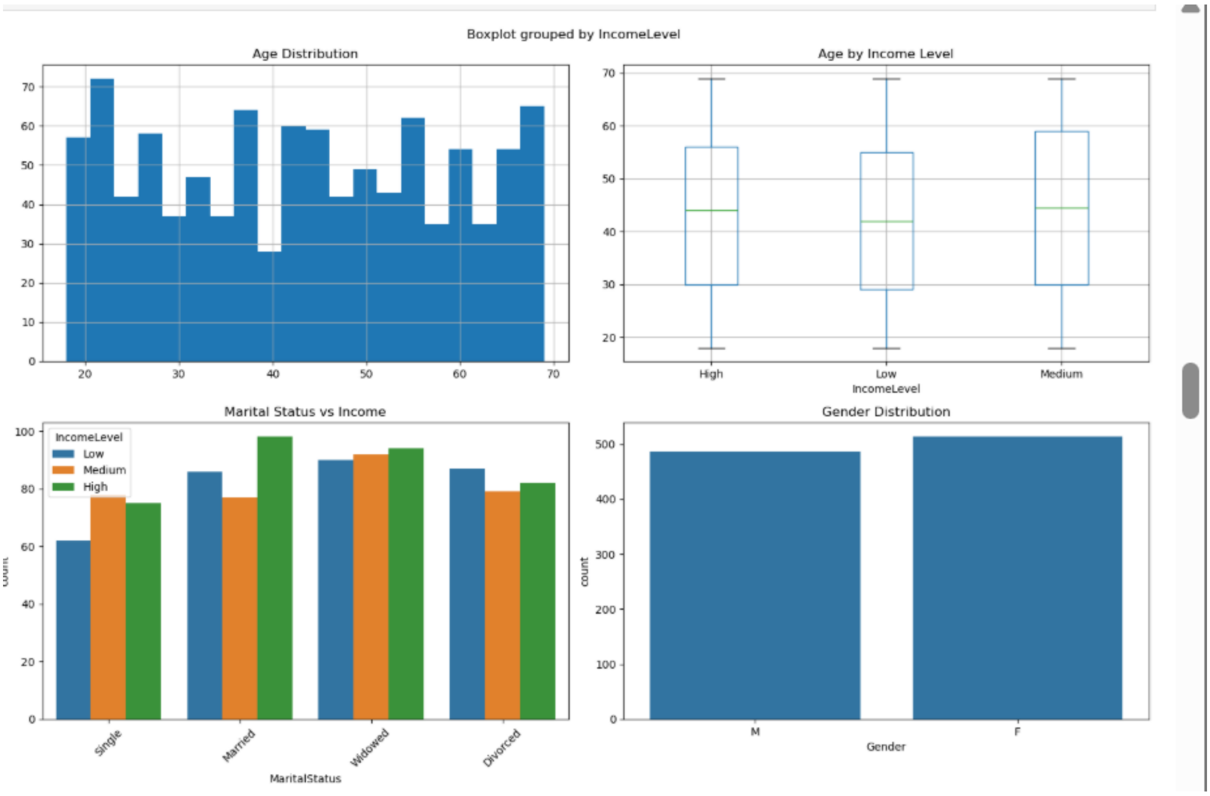


Figure 3: Customer Service Interactions and Churn

- Insights: Service resolution is a critical driver. The plot highlights that churn occurs in both “Resolved” and “Unresolved” cases, but “Unresolved” interactions represent a significant point. Leaving issues unresolves is a known high-risk factor for customer attrition.

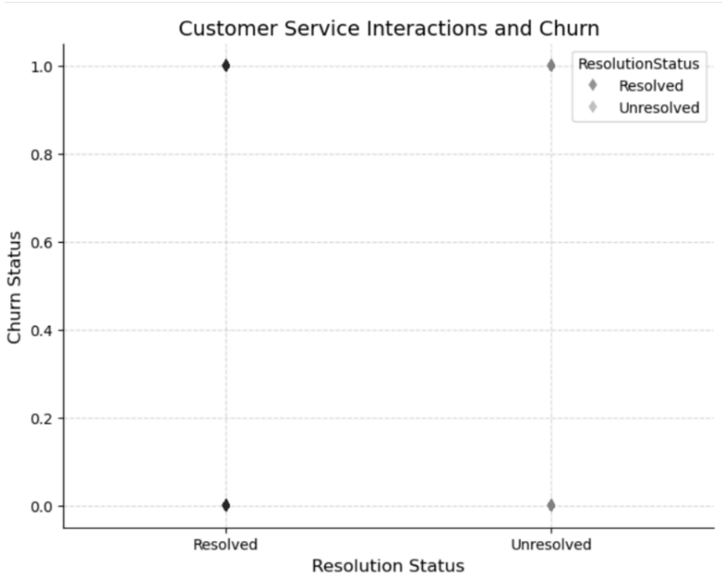


Figure 4: Spending pattern and churn

- Insights: Spending behavior alone does not clearly separate churners from retained customers. Both groups display similar median spending (approx. £250) and distribution shapes. This indicates that high spenders are just as likely to leave as low spenders if other needs (like service) are unmet.

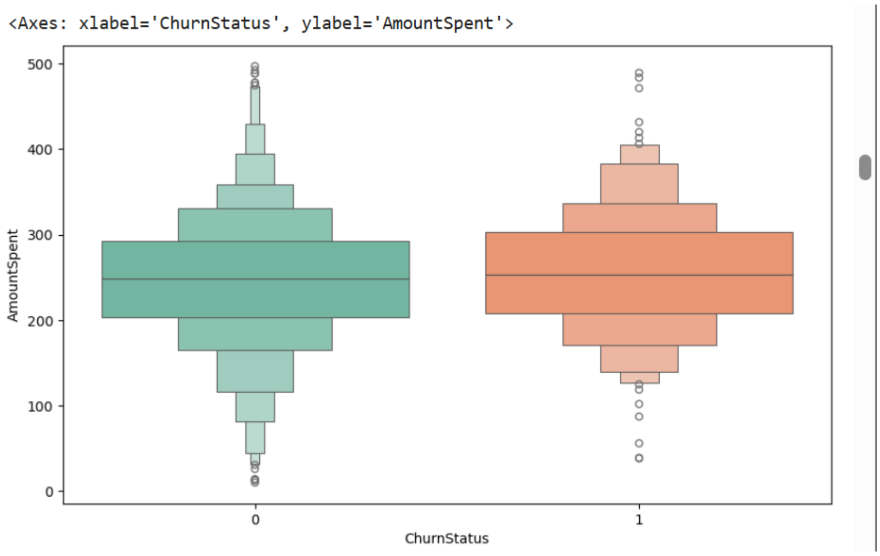


Figure 5: Transaction Patterns

- Insights: Customer activity is uniform. Transaction amounts are evenly distributed between £0-£500, and product popularity (Books, Electronics, etc.) is equal. The "Average Spend per Customer" follows a normal distribution, confirming that the data is clean and free of extreme skews after preprocessing.

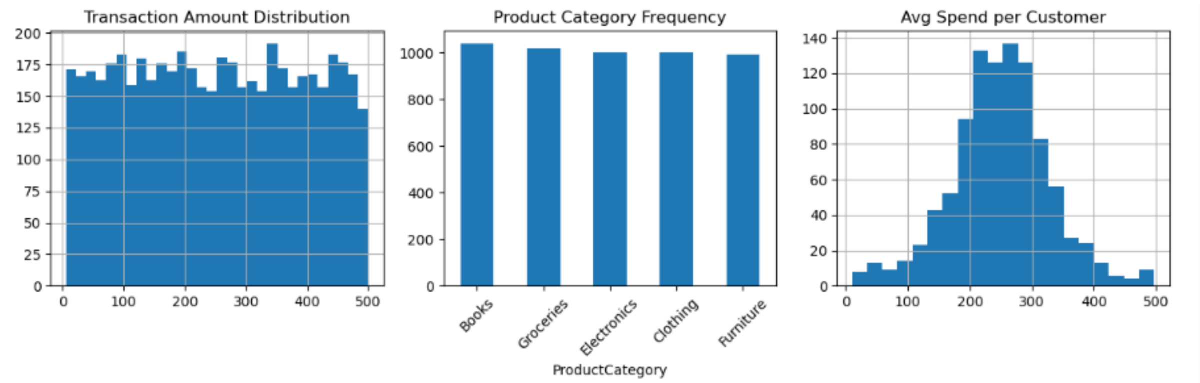
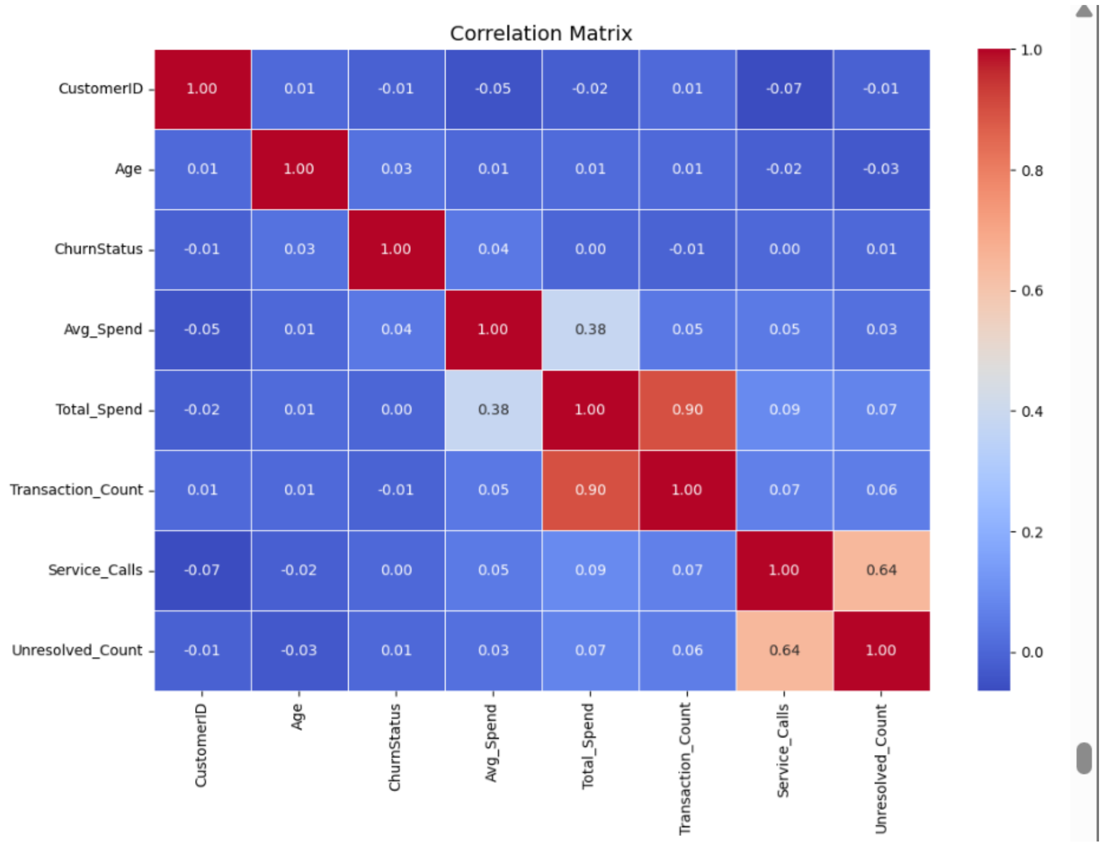


Figure 5: Feature Correlation Matrix

- **Low Target Correlation:** The matrix shows weak linear correlations between individual features and ChurnStatus (all < 0.05). This confirms that churn is driven by complex, non-linear interactions rather than a single factor like "Price" or "Age," validating the need for a machine learning model over simple rule-based logic.
- **Multicollinearity:** We observe a strong correlation (**0.90**) between Total_Spend and Transaction_Count, which is expected (more transactions lead to higher total spend). Similarly, Service_Calls and Unresolved_Count show a moderate correlation (**0.64**). These logical relationships confirm data consistency.



Data Cleaning and Preparation

Data Cleaning and Preparation To ensure model reliability, we performed a rigorous cleaning process:

- **Data Integrity:** Verified that the dataset contained **zero missing values**, confirming high data quality without the need for imputation.
- **Feature Engineering:** Aggregated raw logs into customer-level metrics, creating key features like Avg_Spend and Unresolved_Count.
- **Outlier Treatment:** Capped extreme values in Avg_Spend and Total_Spend at the **99th percentile** to prevent them from skewing the model.
- **Preprocessing:** Applied **One-Hot Encoding** to categorical variables (e.g., Gender) and **Standard Scaling** to numerical variables to ensure all features contribute equally to the analysis.

Model Selection and Building

After testing both XGBoost and Random Forest algorithms, we ultimately selected the **Random Forest Classifier** as our primary model. While XGBoost is a powerful tool, it underperformed in this specific scenario, achieving a lower Recall of 52%.

We refined the Random Forest model using GridSearchCV for hyperparameter tuning and, crucially, adjusted the decision threshold to **0.25**.

- **Why we did this:** We realized that for a retention campaign, missing a churner is far worse than accidentally flagging a happy customer. By lowering the threshold, we created a "safety net" that successfully caught **32 out of 50 (64%)** at-risk customers in our test group.
- **The Trade-off:** This aggressive strategy does lower overall accuracy to 43%, but it ensures the bank doesn't miss the people who matter most—the ones leaving.

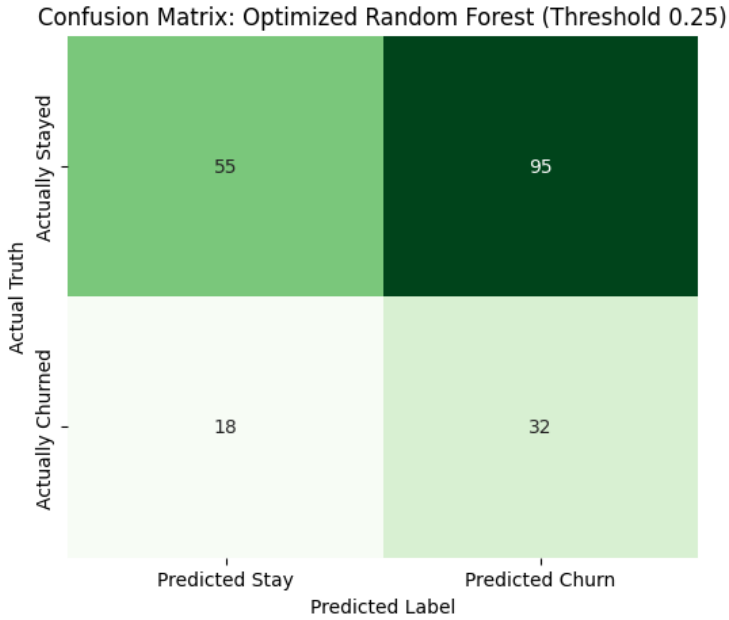
Drivers Analysis of Churn:

- Our analysis revealed that customer churn is driven primarily by behavior, not demographics.
- **Digital Disengagement (LoginFrequency):** This is the strongest warning sign. Customers tend to stop logging in weeks before they actually leave. This "digital silence" is a massive opportunity for early intervention.
- **High-Value Risk (Avg_Spend):** Surprisingly, our highest spending customers are also the most likely to leave. This suggests they are sensitive to service quality and have other banking options, making them a volatile segment.

Model Evaluation

The model's performance was optimized by adjusting the decision threshold to **0.25** (down from the standard 0.50). We made this strategic choice because, in customer retention, missing a churner is far more costly than accidentally flagging a happy customer. We effectively built a "safety net" to catch as many at-risk clients as possible.

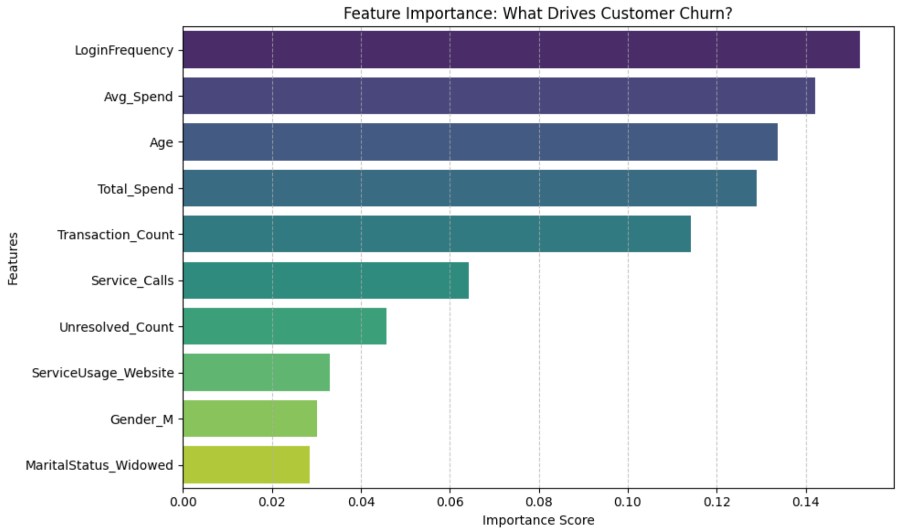
Performance: This approach achieved a **Recall of 64%,** successfully identifying **32 out of 50** churners in the test set. In comparison, baseline models typically captured less than 10%, meaning our tuned model is significantly more effective at preventing silent attrition.



Driver Analysis

The Feature Importance Analysis (Figure 1) told a clear story: churn is driven by **silence**, not noise.

- **Digital Disengagement:** LoginFrequency emerged as the primary predictor. This confirms that customers stop engaging with the app or website long before they close their accounts.
- **The "Silent Churn" Insight:** Notably, Service_Calls ranked significantly lower in importance. This indicates that most churners do not call to complain before leaving they simply disengage and drift away.



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Recommendations and Business Implications

The insights from this model allow the bank to shift from a reactive stance (waiting for complaints) to a proactive one (preventing departures). Based on our findings, we propose the following strategic actions:

1. Implement a 'Silent Churn' Trigger Since LoginFrequency is the top predictor, we know that customers disengage digitally before they leave financially. We recommend implementing an automated alert system: if a customer's login activity drops by **15% month-over-month**, they should automatically receive a personalized re-engagement email ("We missed you!"). This addresses the "silent" risk before the account is actually closed.

2. Prioritize High-Value Protection Our analysis confirmed that Avg_Spend and Total_Spend are critical drivers of churn. This means the bank is at risk of losing its most profitable customers. Retention budgets should be reallocated to focus specifically on the **top 20% of spenders**. Rather than generic discounts, these high-value clients should be targeted with exclusive fee waivers or premium perks to reinforce their loyalty.

3. Future Development While the current Random Forest model provides a robust safety net, future iterations could be enhanced by integrating **competitor pricing data**. This would help the bank understand the external market factors influencing the remaining unexplained churn, further refining our prediction accuracy.